GCE Design & Technology:
Product Design

Smart & Modern Materials Booklet

Name: …… Form: ……

Specification statement:

To apply a knowledge and understanding of **working properties**, **characteristics**, **applications**, **advantages and disadvantages** of the following types of materials in order to discriminate between them and select appropriately.

Smart and modern materials:

a) thermo-ceramics
b) shape memory alloys (SMA)
c) reactive glass
d) liquid crystal displays (LCD)
e) photo-chromic materials
f) thermo-chromic materials
g) quantum tunnelling composites.

1. Using the key terms below, define the term 'smart material' in a few words:

Change, property, significant, reaction, temperature, light

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2. Describe TWO general advantages to the consumer of smart materials-based products

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3. Describe TWO general disadvantages to the consumer of smart materials-based products

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a) Thermo-ceramics

Thermo-ceramics are normally combinations of metals and ceramics which have been manufactured using the sintering process.







Thermo-ceramic properties

Thermo-ceramic Advantages	Thermo-ceramic Disadvantages

Thermo-ceramics uses

b) Shape memory alloys (SMA)

Nitinol
Eye glass frames
Dental arch wires
Orthopaedic implants
Thermal valve







SMA properties

SMA Advantages	SMA Disadvantages

SMA uses

c) Reactive glass

Auto-dimming rear view mirrors,
Welding visors
Office / Apartment privacy windows







Reactive glass properties

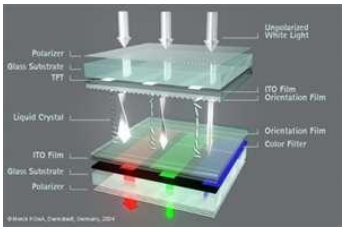
Reactive Glass Advantages	Reactive Glass Disadvantages

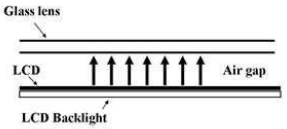
Reactive glass uses

d) Liquid crystal displays (LCD)









Liquid crystal is suspended between layers of glass. The crystal is organised into cells on a matrix that can be electrically charged. In the original designs the crystal darkens and a backlight is needed to see the display in the dark. In more modern examples a filtering system is used to achieve fullcolour representation.

LCD properties

LCD Advantages	LCD Disadvantages

LCD uses

e) Photo-chromic materials

Phosphorescent Pigments







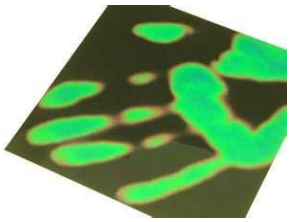
A light gathering pigment that stores the energy until you need it at night. It can be applied as ink, paint, sprayed, brushed, ink jet printed, litho printed and screen-printed.

Phosphorescent pigment properties

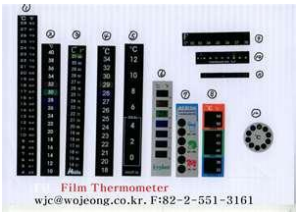
Phosphorescent Pigments Advantages	Phosphorescent Pigments Disadvantages

Phosphorescent pigments uses

f) Thermo-chromic materials









Thermochromic film is a temperature sensitive product that changes its colour according to its ambient temperature. Thermochromic pigment is also used to enhance plastics, for example in the spoons above which change colour according to the temperature of the food.

Thermo-chromic properties

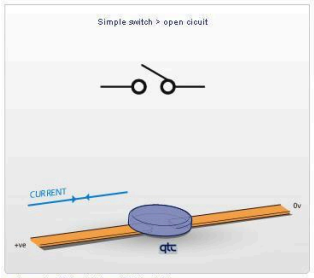
Thermo-chromic pigments advantages	Thermo-chromic pigments disadvantages

Phosphorescent pigments uses

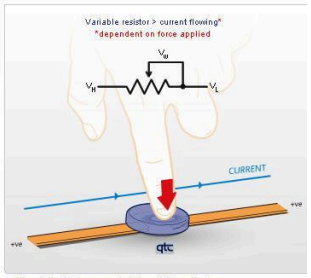
g) Quantum tunnelling composites

Quantum Tunnelling Composite

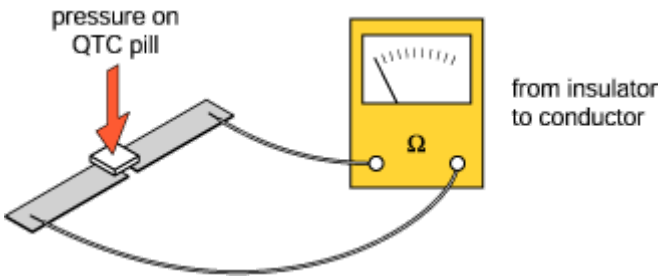
Quantum tunnelling composites (or **QTCs**) are composite materials of metals and non-conducting elastomeric binder (**plastic and rubber**), used as pressure sensors. They utilise quantum tunnelling: without pressure, the conductive elements are too far apart to conduct electricity; when pressure is applied, they move closer and electrons can tunnel through the insulator. The effect is far more pronounced than would be expected from classical (non-quantum) effects alone, as classical electrical resistance is linear (proportional to distance), while quantum tunnelling is exponential with decreasing distance, allowing the resistance to change by a factor of up to 10_{12} between pressured and unpressured states.



▲ Open circuit: Rest state > QTC insulating
In unstressed state, QTC behaves as an insulator ($R>10^{12}$ Ω).



▲ Closed circuit: Compressed state > QTC conducting
Under compression, tension or torsion the resistance drops gradually and controllably. QTC can be capable of passing high currents.



QTC properties

Quantum tunnelling composite advantages	Quantum tunnelling composite disadvantages

Quantum tunnelling composite uses

Quantum Tunnelling Composite Questions

1. Describe TWO features of a Quantum Tunnelling Composites (4)

2. State TWO advantages of Quantum Tunnelling Composites (2)

3. State TWO disadvantages of Quantum Tunnelling Composites (2)

4. Explain TWO Quantum Tunnelling Composite applications in micro-electronics (4)

5. Explain how Quantum Tunnelling Composite use impacts on the manufacture of products (2)